



Team 1 – Attack Surface Management (ASM) Continuous Threat Exposure Management (CTEM) System

1. Introduction

Attack Surface Management (ASM) is the first and one of the most important phases in the CTEM system. The goal of Team 1 is to continuously discover, monitor, and inventory all internet-facing assets of an organization.

This phase helps the organization identify:

- Exposed domains and subdomains
- Live IP addresses
- Open ports and running services
- Technologies used by applications
- Publicly exposed assets that may become attack vectors

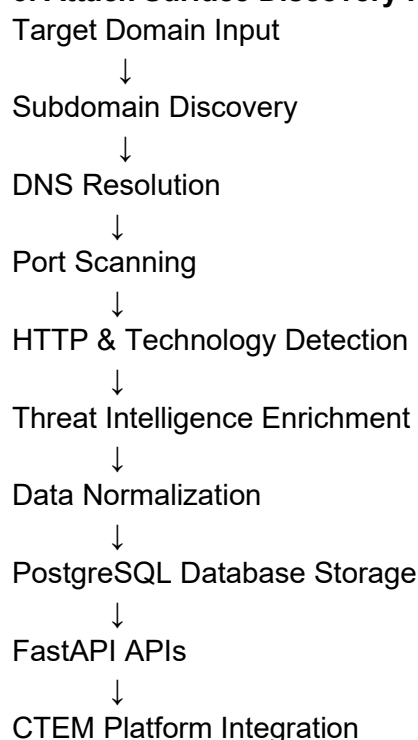
The output generated by Team 1 becomes the foundation for vulnerability analysis, risk scoring, attack path visualization, and monitoring in later CTEM phases.

2. Objective

The main objective of Team 1 is:

- Continuous discovery of internet-facing assets
- Asset inventory creation
- Passive and active reconnaissance
- Monitoring changes in exposed infrastructure
- Preparing normalized asset data for other CTEM modules

3. Attack Surface Discovery Flow





4. Architecture Overview

The ASM module starts with seed inputs such as domains, IP ranges, ASN numbers, GitHub organizations, and cloud assets.

The system then performs:

1. Passive Reconnaissance
2. DNS Resolution
3. Port Discovery
4. Service Enumeration
5. HTTP Fingerprinting
6. Threat Intelligence Enrichment
7. Asset Normalization
8. Database Storage
9. API Integration

All collected data is forwarded to the next CTEM modules for risk analysis and attack path generation.

5. Technology Stack

Programming & Backend

- Python
- FastAPI

Database

- PostgreSQL
- Neo4j (for graph integration)

Reconnaissance Tools

- Subfinder
- Amass
- theHarvester

DNS & Resolution

- dnsx
- MassDNS

Port Scanning & Enumeration

- Naabu
- Nmap
- ZGrab2

HTTP & Web Fingerprinting

- Httpx
- Katana
- waybackurls

Threat Intelligence

- Shodan
- Censys
- VirusTotal

6. Phase-wise Implementation

Phase 1 – Seed Input Collection



The ASM pipeline begins by collecting initial organizational identifiers, which act as the starting point for asset discovery. These inputs help the system identify the digital footprint of the organization.

Inputs Collected

- Domains
- IP ranges
- Email addresses
- GitHub organizations/accounts
- Cloud account information

These seed inputs are stored and processed to initiate the reconnaissance and discovery workflow.

Phase 2 – Passive Reconnaissance

In this phase, the system gathers publicly available information without directly interacting with the target infrastructure. Passive reconnaissance helps identify external assets safely and efficiently.

Tools Used

- Subfinder
- Amass
- crt.sh
- Shodan

Activities Performed

- Subdomain enumeration
- Public asset discovery
- Certificate transparency analysis
- Organizational footprint mapping

Outcome

The system generates a list of discovered hosts, domains, and publicly exposed assets for further analysis.

Phase 3 – DNS Resolution & Validation

After discovering potential assets, the system resolves domain names into live IP addresses to validate active infrastructure.

Tools Used

- dnsx
- MassDNS

Activities Performed

- DNS record resolution
- IP address mapping
- CNAME detection
- Live host validation
- Wildcard filtering

Outcome

The system identifies valid and active hosts connected to the organization.



Phase 4 – Port Scanning & Service Discovery

This phase identifies exposed ports and running services on discovered systems. Open ports help determine which services are publicly accessible and may become potential attack vectors.

Tools Used

- Naabu
- Nmap

Activities Performed

- Open port scanning
- Service discovery
- Version identification
- Network service enumeration

Outcome

The system creates an inventory of exposed services and running applications across the attack surface.

Phase 5 – HTTP & Technology Detection

In this phase, the system analyzes web-based assets and identifies the technologies running behind them.

Tools Used

- Httpx
- Katana

Activities Performed

- HTTP probing
- Technology fingerprinting
- Web application discovery
- Endpoint crawling
- Screenshot collection of web applications

Outcome

The system identifies live web applications, frameworks, technologies, and exposed endpoints.

Phase 6 – Threat Intelligence Enrichment

The discovered assets are enriched using external threat intelligence platforms to improve visibility and risk understanding.

Tools Used

- Shodan
- Censys
- VirusTotal

Activities Performed

- Exposure validation
- Reputation analysis
- Threat intelligence mapping
- Vulnerable service identification

Outcome



The system enhances asset information with external intelligence and exposure insights.

7. Final Outcome of Team 1

After implementation, the ASM module continuously discovers and tracks all external assets of an organization.

The system can:

- Detect newly exposed services
- Monitor infrastructure changes
- Identify attack vectors
- Provide real-time visibility of internet-facing assets
- Supply accurate asset data to the CTEM platform

This creates a strong foundation for proactive cybersecurity and continuous threat exposure management.

8. Conclusion

Team 1 plays a critical role in the CTEM architecture because every security analysis depends on accurate asset discovery.

The Attack Surface Management module ensures:

- Continuous visibility
- Real-time asset inventory
- Faster exposure detection
- Better risk prioritization
- Improved organizational security posture

By integrating reconnaissance, monitoring, enrichment, and automation, Team 1 enables the CTEM system to operate proactively rather than reactively.